A Thought: Magnetism as Energy in the Tree Trunk

Finite-Core Dynamo Hypothesis

Slowik, Marcin
Independent Researcher, Poland
ORCID: 0009-0004-6707-4546

Abstract

In my previous works I proposed that extreme gravitational collapse does not create a singularity but a finite high-density core. In this note I extend my own hypothesis: this finite core can have two parts - inner dense part and outer conductive shell with electric currents. Part of the total energy goes to magnetic field. This can explain magnetar-scale fields of 10^15 G as a natural consequence of the finite-core structure, requiring only 0.00003 percent of the gravitational energy. Stability is given by B_n = 0.

Keywords: finite-core, magnetar, B_n=0, S_n=2F_n, my hypothesis

1. My Tree Model

I use my tree notation from my earlier works:
+F_n = growth up, creation
-F_n = roots down, collapse
B_n = 0 = balance, tree stands
S_n = 2F_n = total energy in the trunk
So far in my works S_n was only gravity. Now I say trunk has current. Part of S_n goes to magnetic field.

2. My Finite-Core Idea

In my works referenced below I proposed that collapse stops at a finite core, not singularity. Now I add structure:
Inner part = very dense matter, holds gravity
Outer shell = less dense but conductive, makes current
Like tree trunk: hard wood inside, sap flowing outside.

3. My Equation

Balance in my language:
S_n = 2F_n = E_core + E_magnetic
B_n = 0 = stability condition
E_magnetic = (B^2 / 2 mu0) * V_shell

Simple calculation in my model:
Take star mass 1.4 Sun, radius 10 km.
Gravitational energy E_grav = G M^2 / R approx 3e46 J.
Shell 1 km thick on 10 km core: volume V_shell approx 1e12 m3.

To get magnetar field B = 1e15 G = 1e11 T:
magnetic energy density = B^2 / 2 mu0 approx 4e27 J/m3
E_mag = density * volume approx 1e40 J
Ratio E_mag / E_grav approx 0.00003 percent

Conclusion: only 0.00003 percent of collapse energy needs to go to current in the shell to get 1e15 G. No new physics needed. Natural from my finite-core.

4. B_n = 0 Holds It

If B grows too large, E_magnetic grows, S_n wants to grow, B_n becomes not zero, tree shakes, must release energy as flash, and returns to B_n = 0. This explains magnetar bursts as re-balancing to B_n = 0. Some collapses produce normal neutron star (small E_mag fraction), some produce magnetar (larger fraction) - depends how much of S_n went to shell current.

5. Summary of My Thought

Magnetar is not different object from neutron star. It is the same finite core, but part of its energy went to current in the shell. Dense center holds gravity. Outer shell makes 1e15 G field. With my balance rule S_n = E_core + E_mag = 2F_n and stability B_n = 0. This is extension of my own finite-core hypothesis from my previous Zenodo records.

References - Only My Own Works

1. Slowik, M. (2026). My finite high-density core hypothesis - original tree model with B_n=0 and S_n=2F_n. Zenodo. DOI: 10.5281/zenodo.22150756
2. Slowik, M. (2026). Gravitational Collapse Beyond the Neutron-Star Limit - finite core from Standard Model matter. My own extension.

This note is an independent extension of my own works above, without using any other author's formalism.

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ORCID: 0009-0004-6707-4546